

SANA'A INTL, Yemen

WMO: 414040

Lat: 15.476N

Lon: 44.220E

Elev: 2206

StdP: 77.48

Time Zone: 3.00 (E03)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.8	5.0	-18.1	1.0	15.9	-15.2	1.3	16.3	7.6	20.6	7.1	20.1	1.1	220	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.9	30.2	12.6	29.9	12.5	29.2	12.4	16.6	22.3	16.1	22.0	15.6	22.0	5.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	14.0	18.3	14.2	13.2	18.4	13.7	12.8	18.4	55.7	22.3	54.0	22.2	52.3	22.0	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-3.6	17.8	1.8	0.7	-4.8	18.3	-5.9	18.8	-6.9	19.2	-8.1	19.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	15.0	16.8	18.5	19.7	21.3	22.9	23.4	22.4	20.9	17.2	15.6	14.1
	DBStd	3.45	1.64	2.00	1.75	1.30	1.06	1.04	1.20	1.17	1.46	1.33	1.57	1.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	430	105	49	19	4	0	0	0	0	1	41	82	130
	CDD10.0	3284	153	190	264	291	352	388	414	385	328	223	168	128
	CDD18.3	673	0	6	24	44	93	138	156	127	79	5	0	0
	CDH23.3	6820	71	241	381	467	893	1414	1336	1051	784	113	46	23
CDH26.7	1467	1	18	36	47	154	426	422	252	112	0	0	0	
Wind	WSAvg	2.5	2.1	2.2	2.4	2.4	2.7	2.8	2.9	2.8	2.7	2.4	2.0	2.0
Precipitation	PrecAvg	189	1	2	25	41	18	8	33	55	4	4	3	1
	PrecMax	364	3	21	108	182	76	83	92	180	27	24	20	5
	PrecMin	68	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	84	1	5	30	39	20	18	24	42	6	6	6	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	28.2	28.9	28.9	30.0	31.2	31.2	30.2	29.9	26.1	25.7	25.1
		MCWB	10.2	11.4	11.8	12.1	12.2	12.9	13.1	13.4	12.8	10.4	10.6	10.7
	2%	DB	25.1	27.1	27.8	28.0	29.0	30.2	30.8	29.7	28.9	25.2	24.5	24.0
		MCWB	9.4	10.6	11.6	11.7	11.8	12.4	12.8	13.3	12.3	10.1	10.6	10.3
	5%	DB	24.1	26.1	26.8	27.1	28.2	29.9	30.0	29.0	28.0	24.8	23.8	23.1
		MCWB	9.6	10.3	11.2	11.6	11.8	12.2	12.6	13.2	12.0	10.0	10.6	9.7
	10%	DB	23.0	25.0	25.8	26.1	27.6	29.1	29.1	28.1	27.1	23.9	22.8	22.0
		MCWB	9.6	10.4	11.2	11.7	11.7	12.7	13.3	13.3	11.7	9.8	10.2	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	14.2	15.7	16.2	16.6	16.1	17.4	17.2	15.9	14.7	14.0	13.9
		MCDB	21.1	22.8	22.9	22.0	21.2	23.4	22.6	22.9	23.0	19.1	18.6	22.2
	2%	WB	12.2	13.2	14.8	15.5	15.7	15.3	16.6	16.6	15.2	13.5	12.8	13.1
		MCDB	20.5	21.8	21.9	21.5	21.6	24.0	22.4	22.2	22.8	18.8	19.8	20.6
	5%	WB	11.6	12.6	14.2	14.8	15.2	14.6	15.9	16.1	14.4	12.0	12.0	12.1
		MCDB	20.2	21.5	21.5	21.0	21.3	24.1	22.2	21.6	22.5	21.5	20.1	19.6
	10%	WB	10.8	11.8	13.5	14.2	14.5	14.0	15.4	15.6	13.6	10.7	11.3	11.0
		MCDB	19.9	21.3	21.0	20.7	21.1	23.8	22.5	21.6	22.6	22.6	20.5	19.3
Mean Daily Temperature Range	5% DB	MDBR	17.2	17.0	15.3	13.8	13.1	12.9	12.1	12.3	13.1	14.1	15.3	17.1
		MCDBR	18.1	17.7	16.3	14.8	13.8	13.6	13.0	12.9	13.3	14.6	16.1	17.5
		MCWBR	8.0	7.2	6.5	5.8	5.2	5.2	4.4	4.1	5.0	6.2	7.1	8.0
	5% WB	MCDBR	15.7	16.1	14.1	13.3	12.7	12.9	12.1	12.4	12.7	11.9	14.1	14.9
		MCWBR	7.2	6.6	5.3	4.9	4.6	5.4	4.2	4.0	5.1	5.7	6.4	6.8
Clear-Sky Solar Irradiance	taub		0.332	0.367	0.434	0.452	0.482	0.572	0.647	0.610	0.499	0.375	0.353	0.332
	taud		2.383	2.253	2.046	2.013	1.913	1.677	1.581	1.655	1.859	2.207	2.287	2.366
	Ebn at Noon		961	938	882	861	823	744	692	726	812	917	929	950
	Edh at Noon		116	138	174	181	197	246	272	255	207	142	126	115
All-Sky Solar Radiation	RadAvg		5.66	6.23	6.68	6.88	7.10	6.92	6.17	6.23	6.51	6.61	5.90	5.54
	RadStd		0.18	0.25	0.33	0.38	0.31	0.30	0.21	0.20	0.22	0.23	0.18	0.22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page